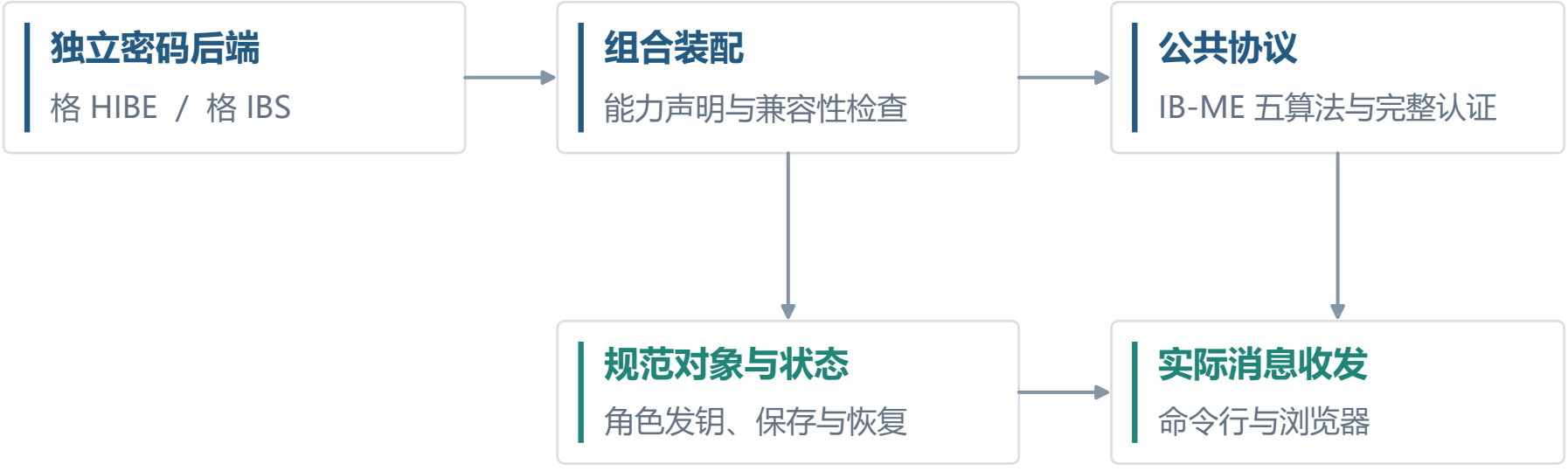从原语实现到完整消息流程